

The Economic Changes in DICE-2023: Plain English Explanation

The Big Picture

The economic side of the DICE model basically asks: "How much does climate change hurt the economy, and how much should we spend to prevent it?" Between 2013 and 2023, the model was updated to reflect that climate change is **more expensive than we thought** and that preventing it is **also more expensive than we thought** - but the damages went up MORE than the costs, so the model now says we should do more to stop climate change.

1. CLIMATE DAMAGES (How much does warming hurt GDP?)

DICE-2013:

- If the world warmed by 3°C, the model said we'd lose about **1.2% of global GDP**
- This was based on older studies from before 2013

DICE-2023:

- If the world warms by 3°C, the model now says we'll lose **3.1% of global GDP**
- This is **2.6 times higher** than before!

Why the change? The 2023 version includes three things:

1. **More recent studies** (56 estimates from 33 papers published through 2023)
2. **Climate tipping points** (like ice sheet collapse or Amazon dieback) - adds 1% GDP loss
3. **Stuff we're definitely missing** (like social unrest, migration, biodiversity loss) - adds another 0.5% GDP loss

Plain English: We now think climate change will hurt the economy a lot more than we thought 10 years ago. At 3°C warming, we'd lose about \$3 trillion per year instead of \$1 trillion per year (in today's economy).

2. ABATEMENT COSTS (How much does it cost to reduce emissions?)

DICE-2013:

- Used theoretical estimates of what it would cost to go to zero emissions
- Assumed costs would fall slowly over time
- Only counted the cost of reducing industrial CO₂

DICE-2023:

- Used **actual data from detailed energy models** that map out realistic pathways
- The "backstop price" (cost to eliminate all emissions) is **\$515/ton of CO₂ in 2050**
- Costs fall faster early on (1% per year until 2050), then slower (0.1% per year after)
- Counts the cost of reducing **ALL greenhouse gases** - CO₂, methane, nitrous oxide, etc.

The key numbers:

- To get to zero emissions today would cost **11% of GDP**
- By 2100, it would only cost **2.7% of GDP**
- BUT because they now include methane and other gases, the total job is about **40% bigger** than just doing CO₂

Plain English: The 2023 model is more realistic about what technologies actually cost based on real energy system models. Going to zero emissions is expensive today but gets cheaper over time. Also, we now count ALL the gases we need to cut, not just CO₂ from burning fossil fuels.

3. DISCOUNTING (How much do we value the future?)

This is the most technical change but really important for policy.

DICE-2013:

- Used a simple discount rate of about 4% per year
- This means \$100 of damage in 25 years is worth about \$37 today
- Basically treated climate investments like any other investment

DICE-2023:

- Uses a **risk-adjusted** discount rate that starts at 4.5% and **declines over time**

- Accounts for two new things:
 1. **Uncertainty about economic growth** (maybe the future will be poorer than we think)
 2. **Climate investments might be safer** than regular investments (climate beta = 0.5)

The formula is now:

Discount rate = Base rate (1.5%)
- Adjustment for growth uncertainty
+ Adjustment for investment risk (2.5%)

Why this matters:

- The declining discount rate means damages far in the future count for **MORE** than they used to
- This is part of why the model now recommends stronger action
- Near-term rate is 4.5%, but it falls over time, so century-scale damages matter more

Plain English: The 2023 model takes uncertainty more seriously. When the future is uncertain, we should be more cautious. Also, climate investments might actually be **LESS** risky than other investments because they pay off when the economy is struggling (like during climate disasters), so we should value them more.

4. ECONOMIC PRODUCTION & GROWTH

DICE-2013:

- Used population and productivity growth estimates from around 2010
- Basic Cobb-Douglas production function (pretty standard economics)

DICE-2023:

- **Updated all the baseline projections** for population and economic growth
- Used more recent studies about long-term growth rates
- Production function stayed the same (still Cobb-Douglas)
- Time steps are now always 5 years (used to be 10 years early on)

Plain English: They updated their assumptions about how big the economy will be in the future based on newer demographic and economic forecasts. The basic economic framework didn't change - it's still the same standard growth model taught in economics classes.

5. THE SOCIAL COST OF CARBON (SCC)

This is the most important number that comes out of the economic model.

DICE-2013:

- The SCC in 2020 was estimated at about **\$44 per ton of CO₂** (in 2015 dollars)
- This is what each ton of emissions costs society in future damages

DICE-2023:

- The SCC in 2020 is now estimated at **\$66 per ton of CO₂** (in 2019 dollars)
- After adjusting for inflation, this is a **real increase of about 30-40%**

Why it went up:

1. Higher damages (the 2.6x increase)
2. Better carbon cycle (emissions stay in the atmosphere longer than we thought)
3. Risk-adjusted discounting (we value future damages more)

Plain English: Every ton of CO₂ we emit causes about \$66 worth of economic damage over time. This should be the minimum carbon tax or price. The number went up because we now think emissions cause more damage and last longer in the atmosphere.

6. WHAT THE MODEL NOW SAYS WE SHOULD DO

DICE-2013 Recommendation:

- Optimal policy would limit warming to about **3.5°C** by 2100
- This balanced the costs and benefits
- The 2°C target seemed very expensive compared to the optimum

DICE-2023 Recommendation:

- Optimal policy would limit warming to about **2.6°C** by 2100
- The 2°C target is now **much closer to optimal** and easier to justify
- We should cut emissions by about **39% by 2050** (if doing the optimal policy)

- To hit 2°C, we'd need to cut **55% by 2050**

Economic value of acting:

- Following the optimal policy would increase global wealth by **\$120 trillion** (present value)
- That's roughly equal to one year of current global GDP

Plain English: Because damages are higher, the model now says we should do a lot more to fight climate change. The famous 2°C target from the Paris Agreement, which used to look expensive, now looks almost economically optimal. We're talking about \$120 trillion in economic benefits from getting this right.

7. WHAT STAYED THE SAME?

Important things that DIDN'T change:

- **Production function** - still standard Cobb-Douglas (capital and labor)
- **Savings and investment** - still based on optimizing consumption over time
- **Pure time preference rate** - still 1.5% (the base discount rate)
- **Consumption elasticity** - still 1.45 (how much we care about inequality across time)
- **Basic model structure** - still a Ramsey optimal growth model

Plain English: The core economic framework is still the same textbook model. They didn't change the basic economics - they just updated the numbers for damages and costs, and improved how they handle risk and uncertainty.

THE BOTTOM LINE

In 2013, the model said:

- "Climate change will hurt, but not catastrophically"
- "Preventing it is expensive"
- "Do something, but don't go crazy - 3.5°C is fine"
- "Carbon costs \$44/ton"

In 2023, the model says:

- "Climate change will hurt a LOT more than we thought" (2.6x worse)
- "Preventing it is still expensive, but damages went up MORE than costs"
- "We should do a lot more - aim for 2.6°C, and 2°C isn't crazy"
- "Carbon costs \$66/ton, so carbon taxes or prices should be higher"

The key economic insight: The cost-benefit analysis shifted toward more aggressive climate action NOT because fighting climate change got cheaper, but because the economic damages from climate change are now understood to be much larger than we thought a decade ago.

Why Does This Matter for Policy?

If you're setting a carbon tax, designing climate regulations, or deciding whether to invest in clean energy:

1. **The social cost of carbon went from \$44 to \$66** - this should be your floor for carbon pricing
2. **The optimal temperature target dropped from 3.5°C to 2.6°C** - we should aim for less warming
3. **The 2°C target now passes cost-benefit analysis** - it's not just a political target anymore
4. **The stakes are huge** - \$120 trillion in potential economic gains from optimal policy

The model is basically saying: "Yes, preventing climate change is expensive. But letting it happen is MORE expensive. We should be willing to pay more now to prevent larger economic losses later."